Supplementary Material for "Certifying the Outcome of Autonomous Vehicle Fallback Triggers: Closed-Loop Evaluation Under Latency and Dataset Shift"

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This document contains the exploratory and post hoc analyses summarized in Section V of the main paper. All numbers use the same settings as the main paper ($\alpha = 0.05$, $\delta = 0.10$, 1,994 Argoverse 2 calibration and 1,996 test scenarios, 1,504 nuScenes scenarios) and can be regenerated from the released per-scenario records without re-simulation.

S1. Class-Conditional Certification (Post Hoc)

Table S1 lists the differences between the closest prior work and this study.

TABLE S1
LUO *et al.* VERSUS THIS STUDY.

	Luo *et al.*	This paper
Guarantee	$\Pr[\text{no alert} \mid \text{unsafe}] \leq \epsilon$, class-conditional	$\Pr[\text{harm not pre-empted}] \leq \alpha$ over scenarios, high probability
Unsafe label	safety score of logged future $< f_0$	nuPlan harm in the no-fallback run
Predictor	Trajectron++	AutoBot
Alert consequence	none (scored on logs)	MRM executed in closed loop
Cost of alerting	false-positive rate	unnecessary stops, induced collisions, residual harm
Data	nuScenes, Lyft	Argoverse 2, nuScenes, Waymo
Shift	label shift only	dataset shift, weighted repair
Latency	not studied	studied

Table S2 implements the certificate of Luo *et al.* (cited in the main paper) in our framework: Mondrian split conformal prediction on the unsafe class, where unsafe means harm in the reference run and an alert means firing before it, with the finite-sample correction and without their tie randomization. At $\epsilon = 0.05$ on the 316 unsafe calibration scenarios it selects a threshold equivalent to always firing, so the class-conditional target is vacuous for this predictor at this level. The score separates harmful from harmless scenarios well enough for a marginal target (AUROC 0.83) but not for a 5 % miss rate among harmful ones. At the marginal-matched $\epsilon = \alpha/\pi = 0.32$ it selects the same threshold as empirical tuning. Deployed on nuScenes, that certificate has a conditional miss rate of 66 % against its 32 % target. The harm base rate rises from 15.8 % to 22.1 %, but the conditional miss rate is far above what that alone explains, so the class-conditional guarantee does not survive the dataset shift either.

S2. Reverse Shift (Exploratory, Post Hoc)

We also certified on half of the nuScenes scenarios and deployed on Argoverse 2 test, a direction pre-registered as a reverse-shift analysis. As pre-registered, with all scenarios, the procedure *refused to certify* on both halves, because of the 5.0 % floor. On the avoidable subset (post hoc), thresholds certified on nuScenes ($n \approx 715$) give an Argoverse 2 miss rate of 1.9 % and 2.0 %, both below the target, at 40.6 % and 39.5 % unnecessary stops. A certificate from the harder dataset therefore transfers safely but conservatively, while the reverse does not transfer. This asymmetry is a single-pair observation pending replication.

TABLE S2
MARGINAL CERTIFICATE (LTT) VERSUS CLASS-CONDITIONAL CERTIFICATES (LUO-STYLE), THRESHOLDS CERTIFIED ON ARGOVERSE 2. MISS: MARGINAL MISS RATE; COND.: MISS RATE AMONG SCENARIOS WITH HARM IN THE REFERENCE RUN. RATES IN %.

	Argoverse 2 test			nuScenes		
Certificate	Miss	Cond.	Unnec.	Miss	Cond.	Unnec.
LTT, marginal $\alpha = 0.05$	3.7	24.2	28.5	13.2	58.1	26.1
Luo-style, $\epsilon = 0.05$	0.0	0.0	84.9	5.0	19.6	77.9
Luo-style, $\epsilon = 0.32$	4.3	28.5	26.0	14.9	66.0	21.3

S3. Recalibration With a Few Labeled Target Scenes

Table S3 draws k labeled target scenes at random and evaluates on the rest. The nuScenes analysis (avoidable subset, 60 draws) was exploratory and not pre-registered. Its Waymo counterpart (all scenarios, 200 draws) was pre-registered as R1–R4 before any Waymo outcome was read, and its outcomes are reported in Section V of the main paper. On nuScenes, uncertified tuning on 40–60 target scenes exceeds the target in about two draws in three. A certified threshold on the same scenes keeps the target (violations 0–10 %) but stops in 53–81 % of scenarios, against 27 % without target labels. A handful of labeled scenes therefore restores a *valid* certificate, not the original operating point. On Waymo, uncertified tuning violates the target in 69.5 % and 59 % of draws with 40 and 60 target scenes, and weighting the Argoverse 2 calibration set with 60 Waymo scenes ($n_{\text{eff}} = 232$) keeps validity at 0 % violation.

S4. Simulator Sensitivity

Three changes to the simulator, each paired by scenario seed against the main run (reactive traffic, 4.0 m/s^2, 2 Hz), leave every conclusion of the main paper unchanged. Non-reactive log-replay traffic raises the harm base rate (17.7 %

TABLE S3
RECALIBRATION WITH $k = 60$ LABELED TARGET SCENES. NUSCENES: EXPLORATORY, AVOIDABLE SUBSET, 60 DRAWS. WAYMO: PRE-REGISTERED (R1–R4), ALL SCENARIOS, 200 DRAWS. VIOL.: FRACTION OF DRAWS WITH TEST MISS $> \alpha$. RATES IN %.

	nuScenes			Waymo		
Method	Miss	Viol.	Unnec.	Miss	Viol.	Unnec.
AV2-certified, no labels	9.2	–	27.1	2.7	0	45.0
Target-only tuned	6.0	70	36.7	6.2	59	29.6
Target-only LTT	0.8	0	72.5	1.5	2.5	58.6
AV2 + 60, weighted LTT	2.8	0	53.3	1.7	0	49.8

versus the main run's 15.1 %, since logged agents cannot react to an ego that deviates from its log) and, with it, unnecessary stops by 7.9 points [95 % CI 5.3, 10.8]. The miss rate itself does not move (paired difference -0.5 points $[-1.3, 0.0]$). A 2.0 m/s^2 comfort deceleration changes neither miss nor stops (paired difference exactly 0 in both), because both alert-level quantities depend only on the reference run and the firing tick (Observation 1 of the main paper), not on how hard the vehicle then brakes. It does lower induced collisions (2.6 % to 1.6 %), the one quantity that does depend on post-fire kinematics. Decisions at 10 Hz on 100 scenarios give a miss rate of 2.9 % [1.0, 8.2], consistent with the main run within an interval widened by the smaller sample.

S5. CERTIFYING RESIDUAL HARM DIRECTLY (POST HOC)

Update. The attempt described here, with a threshold ordered by the monotone alert-level order and a 5 % target, failed. A later amendment (Amendment 11) certified the same quantity at a 7.5 % target with thresholds ordered by out-of-fold estimated risk, and succeeded (Section V-D of the main paper). The failure below is therefore attributable to the target and the order, not to outcome-level certification as such.

The certificate in the main paper bounds the alert indicator $L(X, \lambda)$, not harm in the run the vehicle then drives. We therefore tried to certify *residual harm* $H(X, \lambda)$, the indicator that the nuPlan harm definition fires anywhere in the executed run, before or after the fallback engages. H is not monotone in λ, because firing earlier can itself create harm (4.5 % of fresh calibration scenarios have a non-monotone H), so Conformal Risk Control does not apply by construction. Learn-then-Test needs no monotonicity but is tested on the same fresh Argoverse 2 split as the replication analysis of the main paper ($n_{\text{cal}} = n_{\text{test}} = 904$, target $\alpha_H = 0.05$, $\delta = 0.10$). The comparisons were pre-registered as N1–N5 before running.

TABLE S4
CERTIFYING RESIDUAL HARM ON THE FRESH SPLIT. RATES IN %, WILSON 95 % INTERVALS. VIOL.: FRACTION OF 200 RESPLITS WHOSE TEST RESIDUAL HARM EXCEEDS 5 %. THE SECOND PREDICTOR GIVES THE SAME PICTURE (OUTCOME-LEVEL LTT: 2.8 % HARM, 84.8 % STOPS, 16.0 % VIOLATIONS).

Certificate on	Resid. harm [CI]	Miss	Stops	Viol.
Alert L, LTT (main paper)	5.0 [3.7, 6.6]	3.4	29.0	–
Residual harm H, LTT	2.8 [1.9, 4.1]	–	84.8	15.0
Residual harm H, CRC	3.9 [2.8, 5.3]	–	37.5	45.5

Three results, each contrary to what we pre-registered, follow from Table S4. First, certifying H at 5 % with LTT selects a threshold at which the vehicle fires in essentially every scenario (84.8 % unnecessary stops), and even then it exceeds the target in 15 % of resplits, above δ, so it is not a valid certificate (N1 refuted). Second, the alert-level threshold's residual harm on this fresh split is 5.0 % [3.7, 6.6], so its interval includes the 5 % target, unlike the 6.5 % on the main split (N2 refuted on fresh data; the interval is wide enough that a modest excess is not excluded). Third, CRC violates in 45.5 % of resplits, but LTT also exceeds δ, so a non-monotone-aware calibrator is not by itself sufficient (N3 refuted). A joint certificate on residual harm and induced collisions over threshold and two MRM decelerations (2 and 4 m/s^2) found no configuration other than always firing (N4 refuted). The practical reading is that, with this score and this maneuver, residual harm of 5 % or less is not reachable by tuning the trigger threshold alone at a useful operating point. A different maneuver or score is needed, not a different calibrator.

S6. SHIFT SEVERITY AND THE COST OF THE INTERVENTION (DESCRIPTIVE)

These two tables are computed from the stored records and test no hypothesis.

Shift severity. Table S5 measures how far each target is from the Argoverse 2 calibration set in the seven pre-deployment covariates used for the density ratio. The domain classifier is evaluated by 5-fold cross-validated AUROC (0.5 means indistinguishable), and the Argoverse 2 test split is a null control. nuScenes is much easier to tell apart than Waymo, has a Kish effective sample size of 66 against 288, and a 5.0 % floor of harm that no trigger can pre-empt. The certificate failed on the more distant target and held on the nearer one, which is consistent with shift severity mattering, but two targets cannot establish that, and the floor alone accounts for 5.0 of the 13.2 points missed on nuScenes.

TABLE S5
SHIFT SEVERITY AND THE ARGOVERSE 2-CERTIFIED THRESHOLD ($\lambda = 1.47$). AUROC: CROSS-VALIDATED DOMAIN CLASSIFIER AGAINST THE ARGOVERSE 2 CALIBRATION SET. FLOOR: HARM PRESENT EVEN WHEN THE FALLBACK FIRES AT THE FIRST DECISION TICK. RATES IN %.

Target	n	AUROC	n_{eff}	Harm	Floor	Miss [CI]
AV2 test (control)	1,996	0.50	1,978	15.1	0.0	3.7 [2.9, 4.6]
Waymo	1,000	0.91	288	15.4	0.1	2.7 [1.9, 3.9]
nuScenes	1,504	0.97	66	22.1	5.0	13.2 [11.6, 15.0]

Scenarios from the same scene are not independent. The Wilson intervals in the main paper treat scenarios as independent. The nuScenes scenario identifiers contain the scene name, which allows a check: the 1,504 nuScenes scenarios in the main analysis come from only 37 scenes (41 per scene on average). Resampling whole scenes (cluster bootstrap, 10,000 draws) widens the interval for the miss rate of the Argoverse 2-certified threshold from [11.6, 15.0] to [6.1, 21.0], which still excludes the 5 % target, but only just. A second, largely disjoint set of nuScenes scenarios (1,342 scenarios from 23

scenes, one scene in common, collected for the replication analyses) gives a miss rate of 9.5 % with Wilson interval [8.1, 11.2] and cluster-bootstrap interval [3.9, 16.5], which includes the target. The first-tick floor also differs between the two samples (5.0 % against 0.0 %), so it varies strongly with which scenes are drawn. The direction of the failure therefore replicates, but with a few dozen scenes per sample the size of the nuScenes failure is far less certain than the main-paper intervals suggest. Argoverse 2 and Waymo identifiers are opaque, so the same check, and group-level resplits for the validity analysis, were not possible there.

Where the induced collisions come from. At the certified threshold on Argoverse 2 test, 2.6 % of scenarios [1.9, 3.3] suffer an induced collision. All 129 induced contact events are other vehicles striking the ego, 99 from behind and 30 from the side. They concentrate where the ego is slow or fires early. Scenarios in the two lower terciles of initial ego speed (the lowest starts at standstill, the middle one below 4.6 m/s) have induced rates of 3.6 % [2.4, 5.3] and 3.5 % [2.3, 5.1], against 0.6 % [0.2, 1.5] in the upper tercile, and scenarios in which the fallback fires at or before the median firing tick (tick 5, 2.5 s) have 9.2 % [6.8, 12.4] against 3.0 % [1.7, 5.1] for later firing. These contacts do not look like a follower caught off guard by hard braking. Of the 129, only 3.9 % occur within 2 s of the fallback firing, the median is 8.3 s after it, and 78 % come more than 5 s after it, by which time a vehicle braking at 4 m/s^2 from city speed has long been stationary. The 129 contacts come from only 51 scenarios, and 67 % of them are in scenarios where the ego starts at standstill, so the fallback there changes little about what the ego does. We therefore cannot attribute this cost to the act of stopping. It may instead come from simulated traffic, driven along logged paths with a car-following model, that does not yield to a vehicle which stays stationary in its path for many seconds, which a real driver would also be expected to avoid (released with the code). The 2.6 % induced rate should be read as an upper bound on the cost that stopping itself causes in this simulator.

To grade severity we re-ran the single forced rollout of each of the 51 scenarios with instrumentation (Amendment 9, descriptive). The re-run reproduces the stored induced-contact steps exactly in 34 scenarios and the first induced contact in 49, but later contacts differ in 17, so the contact dynamics after a first impact are not run-to-run reproducible, which is one more way in which the forced-rollout table is exact only up to the first contact. At the first induced contact the ego's speed has a median of 0.6 m/s (27 % below 0.1 m/s and 65 % below 1 m/s), the other vehicle's speed a median of 3.6 m/s, and the speed difference a median of 2.4 m/s (quartiles 1.2 and 5.8, maximum 11.8; this is a scalar speed difference, not a closing speed along the contact axis). Nineteen of the 51 first contacts are lateral and 32 are from behind, a median 6.5 s after firing. At the firing step an agent was in the ego's lane behind it in 39 of the 51 scenarios, at a median gap of 8.3 m and a median speed of 0 m/s. These are therefore low- to moderate-speed contacts with a slow or stationary ego by traffic that keeps moving along its logged path, not hard-braking surprises. The fallback also costs progress: where it fires, the vehicle stops for 7.4 s on average and completes 49 % of the route, against 94 % where it does not. The rates count scenarios, not contacts.

S7. Direct Execution Against the Lookup

The main paper evaluates every trigger by lookup in a table of forced-fire rollouts, which is exact when a run under the trigger coincides with the reference run until it fires (Observation 1). The assumption is measured in the forced runs but was not tested by executing a trigger. This check, pre-registered as Amendment 7 and extended in Amendment 9, does so. It draws two independent random samples of stored Argoverse 2 test scenarios (120 and 140 scenarios, 252 distinct after merging by seed) and runs four geometric triggers live, with the predictor scoring at every decision tick and no score replay or forced-fire table: the certified LTT threshold with zero and with one decision tick of latency, and the thresholds at the 25th and 75th percentile of the calibration peak scores (about 75 % and 25 % of scenarios fire). Each run is scored with the same nuPlan harm definition and compared, scenario by scenario, with the lookup at the same threshold and latency. No direct run failed. (The first draw was first analyzed as a snapshot of 61 scenarios, because the stop signal described in Amendment 7 did not take effect and the job ran to completion. The figures here supersede that snapshot.)

TABLE S6
DIRECT EXECUTION VERSUS LOOKUP, 252 SCENARIOS × 4 TRIGGERS = 1,008 PAIRS. DISAGREEMENT: PAIRS IN WHICH THE TWO DIFFER.

Quantity	Disagree	Wilson 95 % upper bound
Firing step (or no firing)	0 / 1,008	0.4 %
Missed intervention	0 / 1,008	0.4 %
Unnecessary stop	0 / 1,008	0.4 %
Induced collision	0 / 1,008	0.4 %
Harm anywhere in the run	1 / 1,008	0.6 %

Table S6 shows no disagreement in any alert-level quantity or in the induced collisions. The aggregate rates are identical for every trigger (for example 3.6 % missed and 27.8 % unnecessary stops for the certified threshold, and 4.8 % missed with one tick of latency). One pair differs in harm anywhere in the run (seed 1934, certified threshold): the lookup records harm after the fallback and the direct run does not. We did not trace its cause. The re-run of induced collisions in Section S6 shows that contact events after a first impact are not always reproducible from run to run, which is a plausible source. Route completion is identical in 454 of the 1,008 pairs and differs by 0.0002 on average and 0.036 at most. The pre-registered rule, that the lookup is refuted as a substitute if any quantity disagrees in more than 5 % of pairs, is therefore not triggered, and the upper bound on the disagreement rate of the alert-level quantities is 0.4 %. The check covers one dataset, one predictor and one maneuver, and does not test stochastic traffic, for which assumption (A1) would not hold and the lookup would estimate the outcome under a different random draw than the one the trigger would have met. The magnitudes of the pre-fire deviations in the forced runs were not stored, only whether they exceeded 1 cm.

S8. Replication on Fresh Data and a Second Predictor

Table S7 gives the numbers behind the replication analysis in the main paper. Violation frequencies over 200 resplits have Wilson 95 % intervals of [4.6, 12.0] for LTT with the primary predictor and [5.8, 13.8] with the second, so, as in the main split, the point estimates meet $\delta = 0.10$ but their Monte Carlo intervals do not exclude a violation frequency above it. The uncertified baselines (46–51.5 %) are far outside it. The gap between the geometric and the confidence trigger also replicates: with the same target, LTT stops 29.0 % [26.1, 32.0] of scenarios against 67.0 % [63.9, 70.0] for the tuned confidence trigger on the fresh split, and 30.6 % [27.7, 33.7] against 68.3 % [65.1, 71.2] with the second predictor.

TABLE S7
Replication on fresh, disjoint data ($n_{\mathrm{cal}} = n_{\mathrm{test}} = 904$). Violation: fraction of 200 resplits with test miss $> \alpha$.

	Fresh held-out		2nd predictor	
Method	Miss (%)	Viol. (%)	Miss (%)	Viol. (%)
LTT (geometric)	3.4	7.5	3.9	9.0
CRC (geometric)	4.3	46.0	4.6	47.5
Tuned geometric	4.3	51.5	4.6	51.0
T1 confidence	4.8	50.0	4.6	46.0

Ensemble baseline with the second predictor. The replication above lacked the Monte Carlo dropout trigger. We reran only the reference run of every fresh calibration and test scenario with five dropout passes (Amendment 9), reusing the forced-fire outcomes, which do not depend on any score. The re-run reproduced the stored reference (tick count and geometric trace) in 865 of 904 calibration and 867 of 904 test scenarios, and we dropped the others (4.3 % and 4.1 %), so these figures use 865 and 867 scenarios. On the test scenarios the certified geometric trigger misses 3.2 % [2.2, 4.6] with 26.4 % unnecessary stops, the tuned confidence trigger 5.0 % [3.7, 6.6] with 66.2 %, and the tuned ensemble trigger 5.7 % [4.3, 7.4] with 59.6 %. The tuned ensemble trigger therefore misses the 5 % target, which the pre-registered comparison "among triggers that meet the target" did not anticipate, so the rule as written cannot be evaluated for it. Unconditionally, the certified trigger stops 33.2 points less than the ensemble trigger (paired 95 % CI $[-37.5, -29.0]$) and 39.8 points less than the confidence trigger ($[-43.7, -35.8]$), with a lower miss rate than both. The geometric-versus-ensemble gap is thus reproduced with a second predictor, and the open-loop conformal comparison alone remains untested there.

S9. Certifying Under the Deployed Latency

The main paper shows that a certificate calibrated at zero latency fails when the actuation chain is slower. Because the loss at latency ℓ decision ticks is the stored miss indicator at tick $\tau(X) + \ell$, LTT can be run directly on that loss without new simulation (Amendment 8, pre-registered before the run). We certify on the Argoverse 2 calibration set at $\ell \in \{0, 1, 2\}$ ticks (0, 0.5, 1.0 s), deploy at the same latency on the test set, and repeat over the same 200 half-and-half resplits as the main paper.

TABLE S8
LTT calibrated at the deployed latency ("aware") versus the zero-latency certificate deployed at latency ℓ ("naive"), Argoverse 2 test. Miss and stops on the test set, rates in %. Viol.: fraction of 200 resplits with test miss above 5 %, with Wilson 95 % intervals.

ℓ	Naive miss	Naive viol.	Aware miss	Aware stops	Aware viol.
0	3.7	9.5 [6.2, 14.4]	3.7	28.5	9.5 [6.2, 14.4]
1	5.3	92.0 [87.4, 95.0]	3.6	34.4	12.0 [8.2, 17.2]
2	9.1	100 [98.1, 100]	4.1	42.5	11.5 [7.8, 16.7]

The zero-latency certificate is not valid under latency, as claimed. The latency-aware certificate brings the miss rate back below 5 % on the test set, at a cost of 6 and 14 points of unnecessary stops. Its violation frequency, however, is 12.0 % and 11.5 % at one and two ticks, above the $\delta = 0.10$ we pre-registered, so by our own rule the claim that latency-aware certification restores validity was refuted under the uncorrected count. Under the corrected count of Amendment 11 (Section S17), 1.5 % and 0.5 %, the certificate is indeterminate, not violated, and the pre-registered re-reading holds. Three readings are possible and we cannot separate them. The Wilson intervals include 0.10. The statistic counts resplits whose finite test half exceeds 5 %, not resplits whose population risk does, and the zero-latency case is already 9.5 %. And the loss at a positive latency is less smooth in λ, which may weaken the fixed-sequence test in practice. The guarantee itself is unchanged, since the loss is still a fixed function of the scenario and the threshold, but we report the frequency as measured. Across resplits the mean unnecessary-stop rate of the latency-aware certificate is 27.6, 31.8 and 57.0 %, so at two ticks the cost depends strongly on the calibration draw.

S10. Natural Shifts Inside Argoverse 2: Leave-One-City-Out

The main paper's shift evidence rests on two target datasets. Argoverse 2 labels each scenario with one of six cities, which gives six further natural shifts at no simulation cost. For each city we run LTT ($\alpha = 0.05$, $\delta = 0.10$, geometric score) on all stored calibration and test scenarios outside that city (about 3,000 to 3,800 scenarios) and deploy the threshold on the held-out city. We pre-registered (Amendment 9) that the certificate would count as transferring if the held-out miss rate were at most 5 % in at least 80 % of the cities.

TABLE S9
Leave-one-city-out on Argoverse 2 (pooled calibration and test scenarios, $n = 3{,}990$). Harm: harm rate of the held-out city. Miss and stops: held-out city at the threshold certified on the other cities, with Wilson 95 % intervals for the miss rate. Tuned: miss of the uncertified threshold tuned on the other cities. Rates in %.

Held-out city	n	Harm	LTT miss [CI]	LTT stops	Tuned miss
Dearborn	546	8.4	2.0 [1.1, 3.6]	17.0	2.9
Palo Alto	229	12.7	4.4 [2.4, 7.9]	21.0	4.8
Austin	931	14.2	3.7 [2.6, 5.1]	26.7	4.1
Pittsburgh	870	15.4	4.4 [3.2, 5.9]	30.7	5.1
Washington, DC	450	19.3	5.3 [3.6, 7.8]	26.7	6.2
Miami	964	19.7	6.1 [4.8, 7.8]	30.2	7.0

The certificate met the target in four of the six cities (67 %), below the 80 % we required, so the pre-registered claim is refuted. The uncertified threshold met it in three. The two failures, Miami and Washington, DC, are the two cities with the highest harm rates, and in only Miami does the interval exclude 5 %. The failure is not an artifact of holding the city out: the threshold certified on all cities, which saw them, also misses 5.2 % and 5.1 % there. This is what a *marginal* guarantee implies, since the miss rate is the harm rate times the fraction of harmful scenarios left unalerted. Post hoc and descriptive, across these six cities and the two datasets of the main paper, the held-out miss rate and the harm rate of the target have a Spearman correlation of 0.90 ($n = 8$, so this is suggestive only). A shift in the base rate of harm, which an unlabeled-covariate reweighting cannot see, is therefore a plausible common cause of the failures, and it would also explain why the certificate failed on nuScenes (22.1 % harm) but held on Waymo (15.4 %).

S11. Shift in the Traffic Behavior Model

All main results use reactive IDM traffic. To test a different behavior model we deploy the thresholds certified on reactive calibration data on the non-reactive log-replay arm, paired by scenario seed with the reactive test arm (378 scenarios with complete records). The harm base rate is 16.7 % under reactive traffic and 17.7 % under replay. The certified geometric trigger misses 3.7 % [2.2, 6.1] on replay traffic, so the pre-registered claim that it still meets the 5 % target is not refuted. The tuned confidence trigger also meets the target (3.4 %), but at 71.7 % unnecessary stops against 35.4 % for the certified geometric trigger (paired difference -36.2 points, 95 % CI $[-41.8, -30.7]$). The stop-rate advantage therefore persists under a second traffic model, although both are rule-based and neither is a learned agent model.

S12. Spatial-Group Dependence

Scenarios from the same place or drive may be dependent, which would weaken the exchangeability that the guarantee needs. The converted data carry no recording identifier for Argoverse 2 and Waymo, so (Amendment 10, pre-registered) we group scenarios by the ego's initial position: single-linkage components of scenarios whose ego start points lie within 150 m of each other, formed separately for each Argoverse 2 city. This is a proxy, not a recording identifier. It gives 547 groups over the 3,990 pooled Argoverse 2 scenarios (the largest has 649, 16 % of them, the median group is a single scenario) and 664 groups over the 1,000 Waymo scenarios (the largest has 13).

We repeat the main validity analysis with whole groups assigned to calibration or test (200 resplits, calibration share about 53 %). The violation frequency of LTT is 17.0 % (Wilson 95 % interval [12.4, 22.8]) against 9.5 % for random resplits, and the mean test miss rate is 4.2 %. The pre-registered claim that it stays at or below $\delta = 0.10$ is therefore refuted under the uncorrected count (it is indeterminate, at 1.0 %, under the corrected count of Section S17). A post hoc sensitivity at 50 m, which gives 1,796 groups with the largest at 79 scenarios, yields 14.0 % [9.9, 19.5]. Dependence between nearby scenarios is thus one more reason the high-probability statement is weaker in practice than in theory, although random and group-wise splits differ by less than the Monte Carlo error of either would suggest at 200 resplits.

The miss-rate intervals are far less sensitive. A group bootstrap of the certified threshold's miss rate gives [3.0, 4.5] % on the Argoverse 2 test scenarios (360 groups; Wilson [2.9, 4.6]) and [1.7, 3.8] % on Waymo (664 groups; Wilson [1.9, 3.9]). Place-level dependence therefore does not change the point conclusions about the miss rate, including the conservative result on Waymo.

S13. A Second Architecture: Wayformer

Every main result uses AutoBot, and the second predictor of Section S8 is also an AutoBot. To test whether the findings are specific to the architecture, we repeated the full closed-loop campaign (Amendment 10, pre-registered) on the same 904 calibration and 904 test scenarios with a Wayformer forecaster from the same framework ($\text{minADE}_6 = 0.967$, a partly trained checkpoint, the same 2.1 s history and 6 s horizon). A check on identical simulator states found a median 3 s endpoint disagreement of 1.6 m with AutoBot, so the wrapper produces comparable forecasts. T2 and T3 are not repeated.

TABLE S10
Wayformer arm, Argoverse 2 fresh test scenarios ($n = 904$). Viol.: fraction of 200 resplits with test miss above 5 %. Rates in %.

Method	Miss [95 % CI]	Unnec. stops	Viol.
LTT (geometric)	3.1 [2.2, 4.4]	41.8	7.0
Tuned geometric	3.8 [2.7, 5.2]	36.7	48.0
CRC (geometric)	–	–	44.0
Tuned T1 confidence	4.2 [3.1, 5.7]	67.8	53.0

Both parts of the pre-registered claim hold. LTT is valid (7.0 % violations, Wilson interval [4.2, 11.4], and a test miss of 3.1 %), and the certified geometric trigger stops 26.0 points less than the tuned confidence trigger (paired 95 % CI $[-29.8, -22.1]$), which meets the target with 4.2 % missed. The uncertified geometric threshold and CRC violate the target in 48 % and 44 % of resplits, as with AutoBot. The certificate itself and the geometric advantage therefore do not depend on the architecture. The price does depend on the predictor: with Wayformer the certified trigger stops 41.8 % of scenarios, against 26 to 29 % with the two AutoBot models, which is consistent with a less accurate forecaster needing a more conservative threshold to meet the same target.

S14. Physics Baselines, Ablations and Open-Loop Ranking

The main paper's Table I adds a time-to-collision (TTC) trigger, certified with the same procedure. Because the harm label is itself defined by time-to-collision, the TTC trigger is close to the label, so we repeat the comparison under the alternative harm definitions of Section V (post hoc, not pre-registered). The certified TTC trigger needs fewer unnecessary stops than the predictor-based geometric trigger under three of

the four definitions: collision only ($\alpha = 0.01$) 18.8 % against 45.1 %, TTC < 0.5 s 1.6 % against 15.8 %, the headline TTC < 0.95 s 7.1 % against 28.5 %. Under TTC < 1.5 s the certified TTC threshold is very conservative (77.5 % against 34.0 %). A headway-distance trigger is worse than the geometric trigger under all four. The ablations below and the open-loop ranking behind Section V-B follow.

TABLE S11
ABLATIONS ON THE ARGOVERSE 2 TEST SET, CERTIFIED BY LTT.

Variant	Miss (%)	Unnec. stops (%)
Full score, LTT, no latency	3.7	28.5
Route-following ego corridor	0.0	84.9
Width-only isotropic radius	3.6	27.5
Confidence instead of geometry	3.6	75.8
Ensemble spread instead of geometry	3.6	72.3
CRC instead of LTT	4.3	26.0
Empirical tuning instead of LTT	4.3	26.0
+1 decision tick latency (0.5 s)	**5.3**	26.3
+2 decision ticks latency (1.0 s)	**9.1**	23.4

TABLE S12
OPEN-LOOP AUROC VERSUS CLOSED-LOOP UNNECESSARY STOPS AT THE CERTIFIED THRESHOLD, ARGOVERSE 2 TEST.

Score	AUROC	Unnec. stops (%)
Geometric, isotropic radius	0.843	27.5
Geometric, length-aware box	0.832	28.5
Predictor gap	0.817	31.8
Geometric, route corridor	0.577	84.9
Confidence	0.448	75.8
Ensemble spread	0.442	72.3

TABLE S13
LTT (GEOMETRIC) VERSUS THE BETTER OF T1/T2 UNDER ALTERNATIVE HARM DEFINITIONS, ARGOVERSE 2 TEST. UNNECESSARY-STOP RATES IN %; "BEST" IS THE BETTER OF T1/T2 AMONG THOSE MEETING THE MISS TARGET, AND THE PAIRED 95 % CI IS OF THE STOP-RATE DIFFERENCE.

Harm definition	LTT	Best	Which	Diff. [95 % CI]
Collision only, α=0.01	45.1	83.9	T2	-38.8 [-41.4, -36.2]
TTC < 0.5 s	15.8	38.6	T1	-22.7 [-25.5, -20.1]
TTC < 0.95 s (headline)	28.5	68.3	T1	-39.8 [-42.4, -37.2]
TTC < 1.5 s	34.0	68.3	T2	-34.4 [-36.8, -31.9]

S15. COUNTERFACTUAL TRIGGER: SAFE WINDOWS, SHIFT AND PREVALENCE CORRECTION

Safe windows. On the 1,994 training scenarios, 15.8 % have harm in the no-fallback run. Harm is unavoidable (empty safe window) in 0.35 % of scenarios, so almost all harm is preventable by some firing tick. In 12.7 % of scenarios the window is not contiguous (a safe tick, an unsafe tick, then a safe tick again), which is why the outcome loss is certified directly instead of through a union bound on too-late and too-early firing.

Where residual harm comes from. Classifying each scenario with harm in the executed run at the certified threshold (Amendment 11d, descriptive) shows that most of it is firing too late, and little is firing too early (Table S14). The unavoidable share is 0.35 % on the training split (one-sided 90 % upper bound 0.59 %) and 0.34 % on the test pool (0.53 %).

TABLE S14
RESIDUAL HARM OF THE EXECUTED RUN AT THE CERTIFIED THRESHOLD, SPLIT BY CAUSE (% OF ALL SCENARIOS, TEST POOL OF 2,900). ALERT: LEARN-THEN-TEST AT THE ALERT LEVEL. OTHERS: OUTCOME-CERTIFIED AT 7.5 % RESIDUAL HARM.

Threshold	Too late	Too early	Gap	Unavoid.	Total
Geometric, alert level	4.4	0.5	0.8	0.3	5.9
Learned, outcome	4.2	0.4	0.6	0.3	5.5
Geometric, outcome	3.6	0.5	1.0	0.3	5.3
TTC, outcome	4.5	0.1	0.6	0.3	5.5

Prevalence-corrected certificate (refuted). A marginal miss rate is the harm rate times the miss rate among harmful scenarios, and the held-out miss rate tracked the target's harm rate. We therefore estimated each target's harm rate from unlabeled data with black-box shift estimation (the pilot alert as classifier) and certified a class-conditional miss rate at α over the estimate's upper bound (Amendment 11, N-A). The estimate was unreliable (Table S15): the alert rate shifts for reasons other than harm rate, so the label-shift assumption fails. The corrected certificate met the marginal target on 5 of 8 targets, against 6 of 8 for plain LTT, and the pre-registered claim (at least 7 of 8 and more than plain LTT) is refuted.

TABLE S15
BLACK-BOX SHIFT ESTIMATE OF THE TARGET HARM RATE AND MARGINAL MISS AT THE TARGET, UNWEIGHTED LTT VERSUS THE PREVALENCE-CORRECTED CERTIFICATE (PC). RATES IN %.

Target	True harm	Estimate	Upper bound	LTT miss	PC miss
Austin	14.7	8.7	13.4	4.1	4.9
Dearborn	8.4	0.0	0.0	2.2	8.4
Miami	20.0	33.9	38.4	6.1	1.9
Palo Alto	11.6	0.0	6.5	3.9	7.7
Pittsburgh	15.5	25.1	29.5	4.6	2.0
Washington, DC	17.6	21.8	28.1	4.2	2.1
nuScenes	22.1	4.8	9.4	14.0	16.3
Waymo	15.4	46.3	51.7	2.8	0.5

Counterfactual trigger under shift (refuted). Applying the same correction to the outcome target of the learned trigger (tightening 7.5 % by the ratio of the source harm rate to the upper bound) made it meet the target on 4 of 8 targets and refuse to certify on 3, against 6 of 8 without the correction (fails without it: Miami 10.8 %, nuScenes 8.2 %). The pre-registered claim that the correction helps is refuted, for the same reason.

S16. OUTCOME AUDIT WITH AN UPPER CONFIDENCE BOUND

The alert-level threshold (Learn-then-Test on the Argoverse 2 calibration split, 5 % target) was applied to a fresh split of 904 scenarios, where its residual harm is 6.7 % and total harm (with induced collisions) 8.1 %. One-sided 90 % Clopper–Pearson upper bounds are 7.9 % and 9.4 %. The realized harm on each target is compared with the bound in Table S16. The bound covers the realized residual harm on 7

of 9 targets (78 %), below the 90 % we required, so the pre-registered claim is refuted. The two misses are nuScenes, a real shift, and Palo Alto, where 8.0 % against 7.9 % on 253 scenarios is within sampling noise. The audit is valid when the target resembles the audit split and must be recomputed on target-like labeled data otherwise.

TABLE S16
REALIZED RESIDUAL HARM O AND TOTAL HARM O_{tot} AGAINST THE AUDIT UPPER BOUNDS (7.9 % AND 9.4 %). RATES IN %.

Target	O	O_{tot}	Covered
Fresh test (in distribution)	4.8	7.2	yes
Austin	7.0	8.6	yes
Dearborn	4.4	5.1	yes
Miami	6.8	8.3	yes
Palo Alto	8.0	10.0	no
Pittsburgh	6.3	8.4	yes
Washington, DC	6.7	8.1	yes
nuScenes	15.6	16.4	no
Waymo	4.3	8.3	yes

S17. CORRECTED VALIDITY AND THE HYPOTHESIS LEDGER

The violation frequency reported in the main paper counts resplits whose test-half miss exceeds α, which also counts test-sample noise. We also report a corrected count that only counts resplits whose one-sided 95 % Wilson lower bound exceeds α, and call a method *valid* if the uncorrected count is at most δ, *indeterminate* if only the corrected count is, and *violated* otherwise (Amendment 11). Table S17 gives every method. The separation of Learn-then-Test from the uncertified methods survives the correction, whereas the near-boundary cases (latency-aware calibration, group-wise resplits) become indeterminate rather than violated.

TABLE S17
VIOLATION FREQUENCY OVER 200 RESPLITS, UNCORRECTED / CORRECTED, AND VERDICT. RATES IN %.

Method	Unc. / corr.	Verdict
LTT (geometric)	9.5 / 0.0	valid
Tuned geometric	43.5 / 15.0	violated
CRC	41.0 / 12.0	violated
Tuned T1 confidence	50.5 / 14.0	violated
LTT latency-aware, 0.5 s	12.0 / 1.5	indeterminate
LTT latency-aware, 1.0 s	11.5 / 0.5	indeterminate
Zero-latency LTT at 0.5 s	92.0 / 53.0	violated
Zero-latency LTT at 1.0 s	100 / 100	violated
LTT, fresh data	7.5 / 0.5	valid
LTT, second predictor	9.0 / 0.0	valid
LTT, Wayformer	7.0 / 0.0	valid
LTT, spatial groups (150 m)	17.0 / 1.0	indeterminate
TTC, LTT	1.0 / 1.0	valid
CDT (100 resplits)	59.0 / 0.0	indeterminate
ACI (100 resplits)	0.0 / 0.0	valid

Every claim of Amendment 11 and its addendum was fixed before it was run, and the ledger records each verdict.

S18. MECHANISM, REPLICATION AND SHIFT FOR THE COUNTERFACTUAL TRIGGER

Update. The latency comparisons in this section use the 2 Hz arm, where a one-tick delay is 0.5 s and a certification cliff can move a score to always firing. They are superseded by the 10 Hz test of Section S20, which refuted the claim that the learned trigger beats TTC at realistic latencies.

These analyses were pre-registered in Amendment 11c.

Mechanism (claim refuted). To test whether the predictor adds lead time that physics features lack, we trained three learned triggers with delay-shifted labels at a latency of

TABLE S18
AMENDMENT 11 CLAIMS AND VERDICTS.

Claim	Verdict
P1: the predictor-based trigger stops less than a physics TTC trigger	refuted
N-A: prevalence-corrected certificate meets the target on at least 7 of 8 shifts and beats plain LTT	refuted
N-B: latency-aware LTT is not violated at 0.5 and 1.0 s	holds
N-C: the outcome audit bound covers at least 90 % of targets	refuted
CSWC (i): outcome certificate valid at 7.5 %	holds
CSWC (ii): learned trigger stops less than geometric and TTC at zero latency	refuted
CSWC (iii): prevalence correction helps the learned trigger under shift	refuted
11b: learned trigger valid and cheapest at 0.5 and 1.0 s (2 Hz)	holds, withdrawn (S20)
11c M1: the predictor adds lead time beyond physics features	refuted
11c R1: the learned trigger replicates at 0.5 s on a second AutoBot and Wayformer	holds
12 A12-1: counterfactual-label trigger beats a supervised time-to-harm trigger	refuted
12 A12-3: collision-only outcome certification, learned trigger cheapest	refuted (underpowered)
12 A12-4: joint certification of harm and induced collisions at both latencies	refuted
12 A12-6: learned trigger beats TTC, RSS and geometric at 0.3 and 0.5 s at 10 Hz	refuted
13 A13-1: the TTC frontier is the lowest at 0 to 0.5 s, and above RSS and learned at 0.8 and 1.0 s	(a) refuted, (b) holds
13 A13-2: collision-only outcome certification at 2 % with 1,994 calibration scenarios, TTC fewer stops than geometric	refuted (underpowered)
14 A14-1: outcome certificate violation at most δ under spatial-group resplits, all three scores	holds
14 A14-2: outcome certificate valid on the drift-free subset, all three scores	holds
15 A15-1: cluster-aware LTT violation at most δ under spatial-group resplits at 50 m	holds (kish); the strictly valid variant never certifie
15 A15-2: cluster-aware LTT within 10 points of the stops of plain LTT	holds (+9.9)
15 A15-3: cluster-aware LTT valid under random resplits	holds (kish)

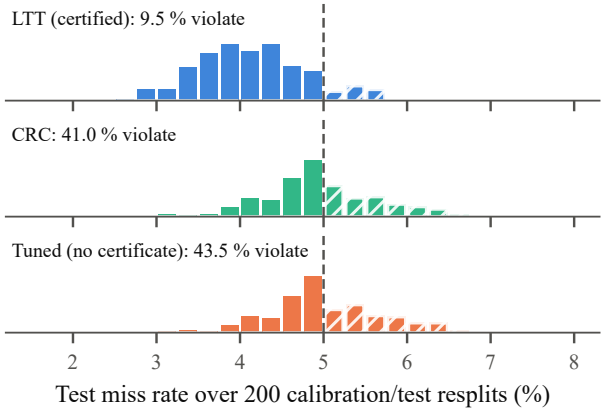


Fig. S1. Test miss rate over 200 calibration/test resplits. Dashed line: the 5 % target; hatched bars: resplits that violate it. A valid certificate keeps violations at or below $\delta = 10$ %. LTT's high-probability guarantee does (9.5 %); CRC's expectation guarantee and empirical tuning do not.

0.5 s (one tick): all features, physics features only (time-to-collision, headway distance, agent count, tick index and scenario covariates) and predictor features only (geometric and confidence scores, predicted gap, tick index and covariates). All three certify the 7.5 % target. Table S19 shows that the full trigger does not stop fewer scenarios than the physics-only trigger (paired CI of the difference $[-0.1, 1.6]$ points), and the predictor-only trigger stops fewer (4.0 points, CI $[2.7, 5.2]$, though it sits closer to the target). The pre-registered claim is therefore refuted: the gain over a plain TTC threshold comes from learning on counterfactual, delay-shifted labels, and either feature group suffices.

TABLE S19
FEATURE-GROUP ABLATION OF THE LEARNED TRIGGER AT 0.5 S OF LATENCY (CALIBRATE 904, TEST 2,900). O: REALIZED RESIDUAL HARM; O_{tot}: WITH INDUCED COLLISIONS. RATES IN %.

Features	O	O_{tot}	Unnec. stops
All	4.9	6.5	25.5
Physics only	4.9	6.6	24.8
Predictor only	5.7	7.3	21.5

Replication (claim holds). With the second AutoBot and with Wayformer the calibration arm is split into 452 training and 452 calibration scenarios and the 904 test scenarios are held out. At 0.5 s the learned trigger meets the target (3.3 % and 4.3 %) and stops 49.3 % and 43.0 % against 84.8 % for both the geometric and the TTC score (paired CIs exclude 0). At zero latency it is worse than the geometric score on the second AutoBot (46.2 % against 39.5 %, CI $[3.1, 10.4]$) and better on Wayformer (44.8 % against 57.0 %); both beat TTC, which only certifies at always firing with 452 calibration scenarios. The stop rates are higher than in the main split because the training and calibration sets are half as large.

Shift (descriptive). Table S20 gives the realized residual harm of the outcome-certified threshold of each score on the eight shift targets of the main paper (zero latency, 7.5 % target). Each score meets the target on 6 of 8. The learned trigger fails Miami and nuScenes, the geometric score nuScenes and Washington, DC, and TTC Miami and nuScenes. No score survives shift better than the others, and TTC is usually cheapest.

TABLE S20
REALIZED RESIDUAL HARM O AND UNNECESSARY STOPS OF THE OUTCOME-CERTIFIED THRESHOLD ON EACH SHIFT TARGET (ZERO LATENCY). RATES IN %. WILSON 95 % INTERVALS FOR THE TTC O, TREATING SCENARIOS AS INDEPENDENT: AUSTIN [3.7, 6.0], DEARBORN [1.9, 4.3], MIAMI [7.8, 10.8], PALO ALTO [2.7, 7.2], PITTSBURGH [4.9, 7.5], WASHINGTON, DC [5.0, 8.8], NUSCENES [7.4, 10.3], WAYMO [3.6, 6.3].

	Learned		Geometric		TTC	
Target	O	Stops	O	Stops	O	Stops
Austin	6.3	8.3	6.6	26.7	4.8	8.6
Dearborn	2.7	7.5	3.5	18.3	2.9	5.3
Miami	10.8	7.0	7.4	35.1	9.2	9.4
Palo Alto	5.1	8.9	5.1	22.9	4.5	6.8
Pittsburgh	5.8	13.0	5.5	37.9	6.1	13.8
Washington, DC	7.4	7.4	7.6	27.1	6.7	8.6
nuScenes	8.2	59.8	12.3	34.8	8.8	20.7
Waymo	4.3	30.4	3.8	48.1	4.8	9.9

S19. OUTCOME PARTITION OF THE FALLBACK

The table partitions every scenario by what happens after the fallback fires, as described in the main paper.

TABLE S21
OUTCOME PARTITION (% OF SCENARIOS). RESIDUAL = MISS + TOO LATE + NEW HARM. INDUCED: NOT-AT-FAULT COLLISION THE NO-FALLBACK RUN AVOIDS.

Data	Trigger	Miss	Too late	Prev.	New harm	Resid. / Induced
AV2	No fallback	15.1	0.0	0.0	0.0	15.1 / 0.0
AV2	LTT (certified)	3.7	1.7	9.8	1.1	6.5 / 2.6
AV2	Tuned geometric	4.3	1.7	9.1	0.9	6.9 / 2.3
AV2	Always fire	0.0	1.0	14.1	3.1	4.1 / 6.5
nuScenes	LTT (AV2-cert.)	13.2	1.0	8.2	1.7	15.6 / 2.4
nuScenes	Always fire	5.0	1.6	16.2	3.8	9.7 / 6.4

S20. AMENDMENT 12: DECISION RATE, SUPERVISED BASELINE AND ROBUSTNESS OF THE LATENCY RESULT

A second strict review asked whether the latency crossover of Sections S9 and S18 survives finer decisions and a simpler learned baseline. Amendment 12 pre-registered the tests. Four of its six claims were refuted, and the crossover is withdrawn as a general finding.

Supervised time-to-harm baseline (refuted). A learned trigger with the same features, model and ordering but a factual label (harm within 2 s after the action takes effect) stops fewer scenarios than the counterfactual-label trigger at every latency (Table S22), so the safe-window label adds nothing as a training target.

TABLE S22
COUNTERFACTUAL-LABEL VERSUS SUPERVISED TIME-TO-HARM TRIGGER, 2 HZ ARM, 7.5 % RESIDUAL-HARM TARGET (CALIBRATE 904, TEST 2,900). STOPS IN %, DIFFERENCE WITH PAIRED 95 % CI (POINTS).

Latency	Counterfactual	Supervised	Difference
0	13.6	9.6	+3.9 [3.0, 4.9]
0.5 s	25.5	23.6	+1.9 [0.7, 3.1]
1.0 s	34.2	29.7	+4.6 [3.4, 5.7]

The certification cliff. The 2 Hz crossover came from where the target sits on the achievable-risk curve. Sweeping the outcome target (Table S23), a 5 % target is unreachable (the learned trigger refuses and the others certify only at always firing), and for TTC at 0.5 s the certified stop rate jumps from 85 % at 7.5 % to 15 % at 9 %, so a small change in the calibration sample moves it across the cliff. At 9 and 10 % the TTC and learned triggers are equal at 0.5 s (15.1 against 14.3 % and 12.4 against 12.0 %).

The 10 Hz arm. For all 904 calibration-split and 904 test-split scenarios of the fresh split we repeated the reference run without the predictor, recording per step the ego speed, the in-lane lead's gap and speed, and the time-to-collision, and ran one forced rollout at every simulator step (10 Hz decision opportunities). The predictor scores of the 2 Hz arm are held at zero order, as a predictor running at 2 Hz would provide, while TTC and the RSS-style safe-distance margin are available at 10 Hz. The new reference runs reproduced the stored per-step TTC traces in 881 of 904 calibration-split and 884 of 904

TABLE S23
UNNECESSARY STOPS (%) OF THE OUTCOME-CERTIFIED THRESHOLD BY TARGET AND LATENCY, 2 HZ ARM. A DASH MARKS A REFUSAL TO CERTIFY; 84.9 IS ALWAYS FIRING. CF.: COUNTERFACTUAL-LABEL LEARNED TRIGGER; G: GEOMETRIC.

	0 s			0.5 s			1.0 s		
Target	Cf.	G	TTC	Cf.	G	TTC	Cf.	G	TTC
0.05	–	84.9	84.9	–	84.9	84.9	–	–	–
0.06	28.2	42.9	84.9	37.9	84.9	84.9	–	84.9	84.9
0.075	13.6	32.8	10.2	25.5	37.5	84.9	34.2	84.9	84.9
0.09	6.2	28.3	7.8	14.3	27.8	15.1	24.5	84.9	84.9
0.10	5.2	22.7	7.1	12.0	25.3	12.4	22.3	39.2	84.9

test-split scenarios (97.5 % and 97.8 %), short of the 99 % we required; the others were dropped. The learned triggers are trained on 440 scenarios, calibrated on 441 and tested on 884, so the stop rates are higher than on the 2 Hz arm for the learned triggers. Table S24 gives realized residual harm and stops. Against the learned counterfactual-label trigger, the paired stop differences (learned minus other) are $+13.2$, $+16.5$, $+18.9$ and $+13.0$ points for TTC at 0, 0.2, 0.3 and 0.5 s (all CIs exclude 0, with $+10.7$ to $+22.1$ at their extremes), so the pre-registered claim that the learned trigger beats TTC, the safe-distance trigger and the geometric trigger at 0.3 and 0.5 s is refuted. At 0.8 and 1.0 s, added post hoc because the forced-rollout table covers every delay, TTC and the geometric trigger certify only at always firing.

TABLE S24
10 HZ ARM: REALIZED RESIDUAL HARM O AND UNNECESSARY STOPS OF THE OUTCOME-CERTIFIED THRESHOLD (7.5 % TARGET). BOTH IN %. LEARNED (CF.) AND (FACT.): COUNTERFACTUAL AND FACTUAL LABELS.

	TTC		Safe dist.		Geometric		Learned cf.		Learned fact.	
Latency	O	Stops	O	Stops	O	Stops	O	Stops	O	Stops
0 s	4.1	5.8	4.9	20.1	3.5	32.9	5.1	19.0	6.4	7.6
0.2 s	5.0	8.7	5.0	24.7	4.9	28.5	5.5	25.2	5.0	17.6
0.3 s	4.6	9.5	5.4	23.9	5.7	26.5	5.9	28.4	4.2	20.4
0.5 s	5.0	12.6	6.6	23.6	5.3	31.9	6.8	25.6	4.2	30.7
0.8 s	4.1	86.7	6.1	29.9	4.1	86.7	4.9	36.5	5.1	31.4
1.0 s	3.8	86.7	5.3	40.2	3.8	86.7	5.9	30.9	4.6	39.6

Collision-only harm (underpowered). With collision-only harm (base rate 4.0 % in the calibration split) and a 2 % outcome target, no score certifies at zero latency and only always firing certifies at 0.5 s with 904 calibration scenarios, so the pre-registered comparison is refuted and uninformative: the Hoeffding–Bentkus test cannot resolve a 2 % target at this sample size.

Joint certification (refuted). Certifying residual harm at 7.5 % and induced collisions at 3 % together succeeds at zero latency for the learned trigger (harm 4.7 %, induced 2.1 %, 18.9 % stops) and for TTC (5.5 %, 1.0 %, 10.2 %) but for no score at 0.5 s, so the claim that the learned trigger certifies jointly at both latencies is refuted.

Multiplicity. The seven stop-difference comparisons of Amendments 11 and 11b that we recomputed with Bonferroni-adjusted bootstrap intervals (level $1 - 0.05/12$) all keep intervals that exclude zero. The refuted comparisons were not recomputed, because widening an interval cannot support a refuted claim.

S21. AMENDMENT 13: CERTIFIABILITY MAP AND COLLISION-ONLY HARM

Amendment 13 was registered before either analysis was run (Section S17 lists the verdicts). *Certifiability map (claim refuted).* Table S25 gives, for every score, latency and target in a grid of nine, the realized residual harm and unnecessary stops of the certified threshold. The frontier of the main paper is the lowest target in a row that is usefully certified, that is, certified with at most 50 % stops and a realized harm within the target. The pre-registered claim (a), that the TTC frontier is the lowest at 0, 0.2, 0.3 and 0.5 s, is refuted: from 0.2 to 0.5 s the geometric or learned trigger reaches a lower target (at 0.2 s only the learned trigger reaches 5 %). Claim (b), that TTC's frontier lies above the lowest of the RSS and learned frontiers at 0.8 and 1.0 s, holds (8 and 12 % against 6 %). *Split dependence (descriptive).* Table S26 shows that the single-split map hides variation: TTC certifies usefully at 7.5 % in 76 % of resplits at 0.5 s, fewer than the geometric (92 %) or RSS (88 %) trigger, and in 18 and 0 % of resplits at 0.8 and 1.0 s.

Collision-only harm (claim refuted). With harm defined by at-fault collisions only, so that no time-to-collision term favors the TTC trigger, doubling the calibration set to 1,994 scenarios did not make a 2 % target resolvable (Table S27): geometric refuses at zero latency and TTC certifies only near always firing. At a 3 % target, reported descriptively, both certify at zero latency and TTC needs 21.7 points fewer stops (95 % CI $[-23.4, -20.0]$), but neither certifies usefully at 0.5 s.

S22. AMENDMENT 14: OUTCOME CERTIFICATE UNDER DEPENDENCE AND WITHOUT DRIFTED SCENARIOS

Amendment 14 was registered before either analysis was run, after the pre-submission audit. The set-up is that of the zero-latency outcome certificate of Section S15 (training on `av2_cal`, calibration on `av2_cal5`, test on the two test arms; 7.5 % target; $\delta = 0.10$). *Spatial groups (claim holds).* Ego start points within 150 m in the same city form 536 groups over the 3804 pooled scenarios, the largest holding 18 %. Resplitting whole groups into calibration and test halves, the uncorrected violation frequency is at most δ for all three scores, although the geometric score sits exactly at the boundary (10.0 %), which is the same score whose alert certificate rises to 17.0 % under spatial resplits. *Drift-free subset (claim holds, with a caveat).* Dropping every scenario with a flagged simulator drift leaves the realized harm at most 5.6 % and the violation frequency at most 7.0 %. Stop rates fall for every score (learned 13.6 to 6.0 %, geometric 32.8 to 27.9 %, TTC 10.2 to 7.2 %), and the learned trigger, which was behind TTC on all scenarios, is then ahead of it. The comparison was not tested and the learned model is refit on the smaller training set, so we read it as showing that the 2 Hz zero-latency ordering of the learned trigger and TTC depends on the flagged scenarios. The 10 Hz arm already drops scenarios that did not reproduce the stored reference run (2.2 to 2.5 %).

S23. AMENDMENT 15: CLUSTER-AWARE LEARN-THEN-TEST

The alert certificate violates its target in 17.0 % of spatial-group resplits at 150 m and 14 % at 50 m (Section S12),

TABLE S25
CERTIFIABILITY MAP ON THE 10 HZ ARM: REALIZED RESIDUAL HARM / UNNECESSARY STOPS (BOTH %) OF THE OUTCOME-CERTIFIED THRESHOLD, BY SCORE, LATENCY AND TARGET (SEED-0 SPLIT; CALIBRATE 441, TEST 884). A DASH: NO THRESHOLD CERTIFIED. PARENTHESES: NOT USEFULLY CERTIFIED (MORE THAN 50 % STOPS OR A REALIZED HARM ABOVE THE TARGET). LEARNED: COUNTERFACTUAL LABELS. TARGETS IN %.

Latency	Score	5	6	7	7.5	8	9	10	12	15
0 s	TTC	–	3.6/7	3.6/7	4.1/6	4.1/6	6.0/5	6.0/5	7.7/3	9.4/1
	RSS	–	3.6/42	4.9/20	4.9/20	5.3/16	6.7/13	7.4/8	8.0/7	9.0/4
	Geometric	–	3.4/35	3.5/33	3.5/33	4.3/30	5.1/27	7.0/21	8.9/13	10.9/7
	Learned	–	4.0/33	5.2/21	5.1/19	6.4/12	6.9/10	8.3/7	9.0/5	11.5/1
0.2 s	TTC	–	3.2/12	5.0/9	5.0/9	5.0/9	6.0/7	6.0/7	8.5/4	8.5/4
	RSS	–	(4.0/54)	(4.0/54)	5.0/25	5.0/25	6.8/15	7.6/12	8.3/8	11.7/3
	Geometric	–	3.8/34	4.5/29	4.9/29	5.3/27	6.8/22	7.7/17	8.9/13	11.2/6
	Learned	3.4/49	4.0/38	5.3/27	5.5/25	5.8/24	7.0/18	7.0/18	9.7/8	11.4/3
0.3 s	TTC	(3.2/87)	(3.2/87)	4.6/10	4.6/10	5.5/8	5.5/8	6.7/7	8.3/6	9.7/4
	RSS	(4.4/56)	(4.4/56)	(4.4/56)	5.4/24	6.0/22	7.0/18	7.8/13	8.3/11	11.0/4
	Geometric	3.7/43	4.3/35	5.3/28	5.7/26	5.7/26	7.0/22	7.7/20	8.8/15	10.6/9
	Learned	(3.8/51)	4.8/36	5.0/32	5.9/28	6.1/27	7.9/17	7.9/17	10.2/8	12.0/3
0.5 s	TTC	–	(3.2/87)	5.0/13	5.0/13	5.0/13	6.1/11	6.6/10	8.5/8	11.8/4
	RSS	–	(4.9/51)	5.5/27	6.6/24	6.6/24	7.1/22	8.0/19	9.2/12	11.4/5
	Geometric	–	4.4/41	5.3/32	5.3/32	6.0/27	6.2/26	7.7/22	9.6/16	11.0/9
	Learned	–	3.7/45	5.9/30	6.8/26	6.9/25	7.1/22	9.0/14	10.3/9	12.3/3
0.8 s	TTC	(4.1/87)	(4.1/87)	(4.1/87)	(4.1/87)	5.0/22	6.6/18	6.9/17	9.3/12	11.9/6
	RSS	(4.1/87)	(4.1/87)	(4.1/87)	6.1/30	7.0/25	6.9/25	8.1/20	8.5/19	11.2/8
	Geometric	(4.1/87)	(4.1/87)	(4.1/87)	(4.1/87)	(4.1/87)	7.6/30	7.6/30	8.8/22	11.2/11
	Learned	(4.1/79)	4.9/41	4.9/37	4.9/37	4.9/37	6.4/30	8.1/19	10.4/9	12.0/5
1 s	TTC	(3.8/87)	(3.8/87)	(3.8/87)	(3.8/87)	(3.8/87)	(3.8/87)	(3.8/87)	9.5/15	11.9/7
	RSS	(3.8/87)	5.1/49	5.3/40	5.3/40	5.3/40	7.2/27	8.1/24	8.5/20	12.4/8
	Geometric	(3.8/87)	(3.8/87)	(3.8/87)	(3.8/87)	(3.8/87)	(3.8/87)	8.8/27	9.7/22	11.7/14
	Learned	–	4.1/44	5.1/34	5.9/31	5.9/31	6.6/28	7.8/21	10.3/10	11.8/5

TABLE S26
SHARE OF 50 RANDOM CALIBRATION/TRAINING RESPLITS OF THE 10 HZ ARM IN WHICH THE 7.5 % OUTCOME TARGET IS USEFULLY CERTIFIED (%).

Latency	TTC	RSS	Geometric
0 s	100	88	100
0.2 s	98	86	98
0.3 s	96	84	96
0.5 s	76	88	92
0.8 s	18	48	6
1 s	0	52	4

TABLE S27
COLLISION-ONLY HARM, 2 HZ ARM, CALIBRATION ON 1,994 SCENARIOS, TEST 2,900. REALIZED HARM / UNNECESSARY STOPS (%) AND THE PAIRED STOP DIFFERENCE TTC MINUS GEOMETRIC (POINTS, 95 % CI).

Target	Latency	TTC	Geometric	Difference
2	0 s	1.2 / 96.7	refused	–
2	0.5 s	refused	refused	–
3	0 s	2.1 / 20.7	2.0 / 42.4	-21.7 [-23.4, -20.0]
3	0.5 s	0.9 / 96.7	0.9 / 96.7	+0.0 [+0.0, +0.0]

because its Hoeffding–Bentkus p-value assumes independent scenarios. The cluster-aware certificate keeps the empirical risk and the fixed-sequence walk but evaluates the p-value at $n_{\text{eff}} = N^2/\sum_g n_g^2$, where n_g are the sizes of the spatial groups in the calibration set. The *kish* variant uses the Hoeffding–Bentkus p-value at $\lfloor n_{\text{eff}} \rfloor$ and is a heuristic. The *Hoeffding* variant uses the weighted Hoeffding bound $\exp(-2n_{\text{eff}}(\alpha - \hat{R})^2)$ for independent group means, which is strictly valid if groups are independent. The effective sample sizes were computed before the registration (Amendment 15), the outcomes after. *At 50 m (claims hold, with caveats).* The kish variant removes

TABLE S28
OUTCOME CERTIFICATE (7.5 % TARGET, ZERO LATENCY, 2 HZ ARM) UNDER RANDOM AND SPATIAL-GROUP RESPLITS AND WITHOUT DRIFTED SCENARIOS. REALIZED TEST HARM AND UNNECESSARY STOPS (%), AND VIOLATION FREQUENCY OVER 200 RESPLITS, UNCORRECTED / CORRECTED (%). THE DRIFT-FREE SETS HOLD 1928/864/1943/878 SCENARIOS (TRAINING, CALIBRATION, TEST AND TEST5 ARMS) AGAINST 1994/904/1996/904.

Set	Score	Harm	Stops	Random resplits	Group resplits
All scenarios	Learned	5.4	13.6	4.5 / 0.5	7.0 / 1.5
	Geometric	5.3	32.8	7.5 / 0.0	10.0 / 0.5
	TTC	5.5	10.2	0.0 / 0.0	2.0 / 0.5
Drift-free	Learned	5.6	6.0	5.0 / 0.5	–
	Geometric	5.1	27.9	7.0 / 0.0	–
	TTC	5.6	7.2	1.0 / 0.5	–

the violations (0.0 % against 14.0 %, also 0.0 % under random resplits) and its mean test miss is 2.3 % against 4.1 %, so it is valid but conservative: it needs 37.5 % unnecessary stops against 27.6 %, a paired difference of 9.9 points, just inside the 10-point limit that was registered. The strictly valid Hoeffding variant refuses in all 200 resplits, because an effective sample of about 226 scenarios cannot reject a 5 % risk at $\delta = 0.10$ for any threshold with a useful stop rate, so the claim for it is vacuous. *At 150 m (descriptive).* One group holds 16 % of the pool and the calibration half has $n_{\text{eff}} \approx 10$, so both variants refuse in every resplit: if dependence extends over that scale, 3,990 scenarios cannot support a certificate. The repair therefore quantifies what spatial dependence does to the certificate rather than removing the problem.

TABLE S29

CLUSTER-AWARE LEARN-THEN-TEST FOR THE ALERT CERTIFICATE (GEOMETRIC SCORE, $\alpha = 0.05$, $\delta = 0.10$) OVER 200 RESPLITS OF THE POOLED 3,990 ARGOVERSE 2 SCENARIOS. GROUP RESPLITS: VIOLATION FREQUENCY UNCORRECTED / CORRECTED (%); MEAN UNNECESSARY STOPS AND TEST MISS (%); RANDOM-RESPLIT VIOLATION (%). n_{eff}: MEAN CLUSTER EFFECTIVE SAMPLE SIZE OF THE CALIBRATION HALF. GROUPS: SINGLE-LINKAGE EGO STARTS WITHIN 50 OR 150 M, PER CITY.

Groups	Certificate	n_{eff}	Group viol.	Stops	Miss	Random viol.
50 m	Plain LTT	1999	14.0 / 0.5	27.6	4.1	8.5
m	Cluster-aware (kish)	226	0.0 / 0.0	37.5	2.3	0.0
m	Cluster-aware (Hoeffding)	226	refuses (200 of 200)	–	–	refuses
150 m	Plain LTT	2117	17.0 / 1.5	27.1	4.2	8.5
m	Cluster-aware (kish)	10	refuses (200 of 200)	–	–	refuses
m	Cluster-aware (Hoeffding)	10	refuses (200 of 200)	–	–	refuses